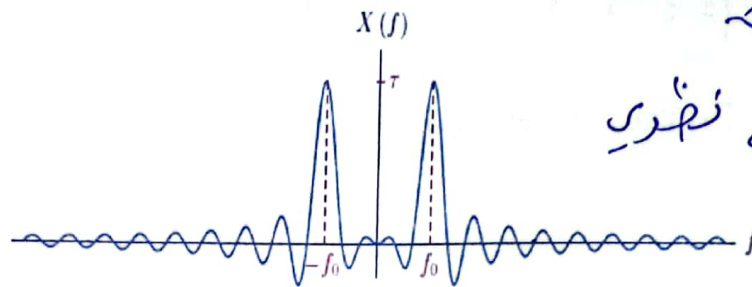


Signals and Systems

Lecture 3 : Operations of Signals



زجاج قاله
اسرار رفيع نظري
سافر ٣

Amjed Jumaah

Basic Operations of Signals

Operation performed on independent variable:

Time Shift

For any t_0 and n_0 , time shift is an operation defined as

$$x(t) \rightarrow x(t - t_0)$$

$$x[n] \rightarrow x[n - n_0]$$

If $t_0 > 0$ the time shift is known as "delay". If $t_0 < 0$ the time shift is known as "advance".

Example. In Fig. given below, the left image shows a continuous-time signal $x(t)$. A time- shifted version $x(t - 2)$ is shown in the right image.

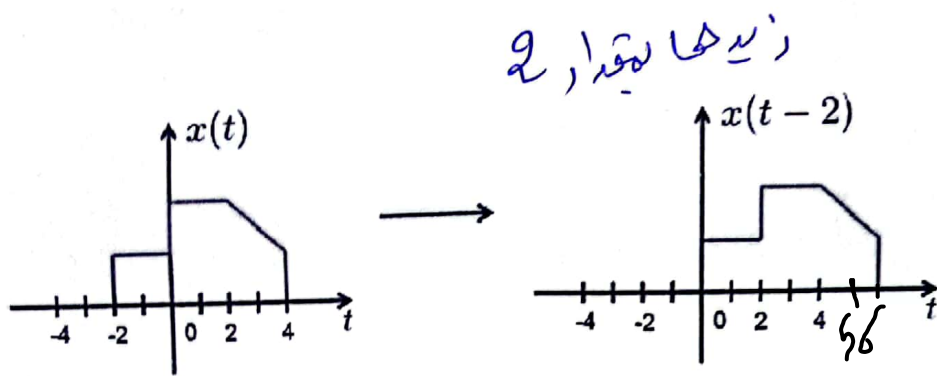
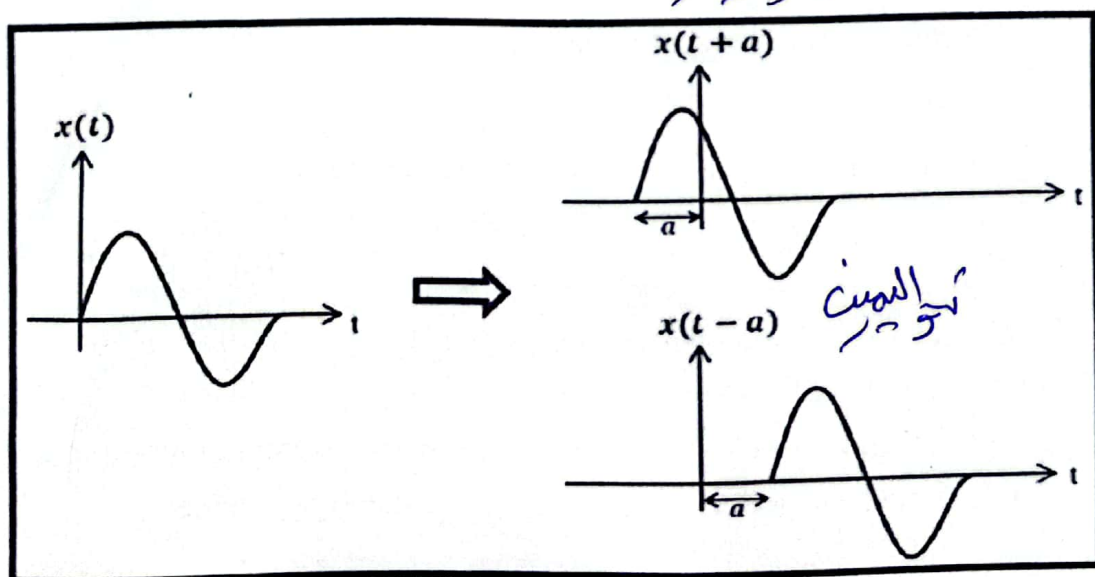


Figure: An example of time shift.

Example 2 نحو اليسار



عكس نقل الوقت

Time Reversal

Time reversal is defined as $x(t) \rightarrow x(-t)$

$$x[n] \rightarrow x[-n],$$

تدوير تناظري

which can be interpreted as the "flip over the y-axis".

Example:

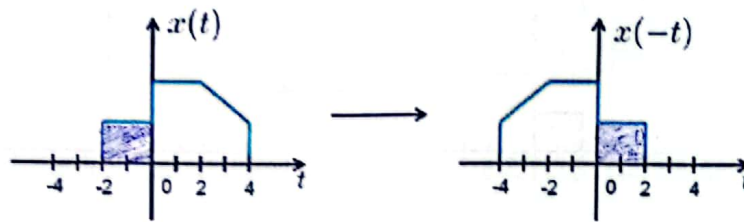
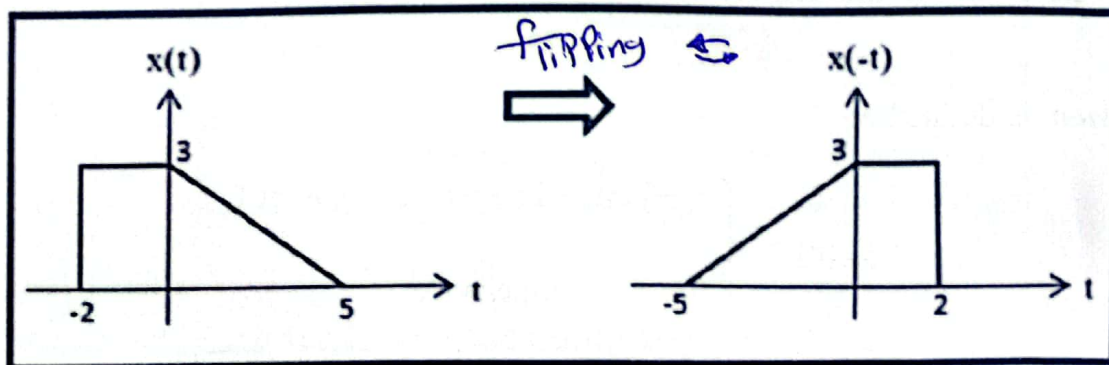


Figure: An example of time reversal.

Figure

Example 2



لحساب

تسريع / تقليل الوقت

Time Scaling

Time scaling is the operation where the time variable (t) is multiplied by a constant a :

$$x(t) \rightarrow x(at), \quad a > 0$$

تسريع / تقليل

If $a > 1$, the time scale of the resultant signal is "decimated" (speed up).

If $0 < a < 1$,

the time scale of the resultant signal is "expanded" (slowed down).

توسيع

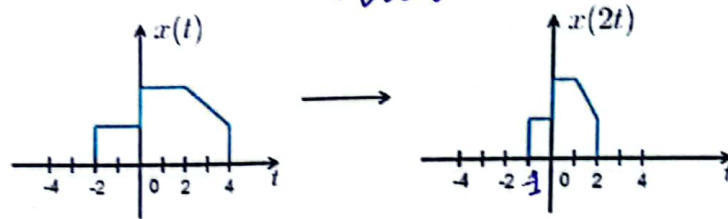


Figure : An example of time scaling.

Decimation and Expansion

Decimation and expansion are standard discrete-time signal processing operations.

Decimation is defined as

نفس a

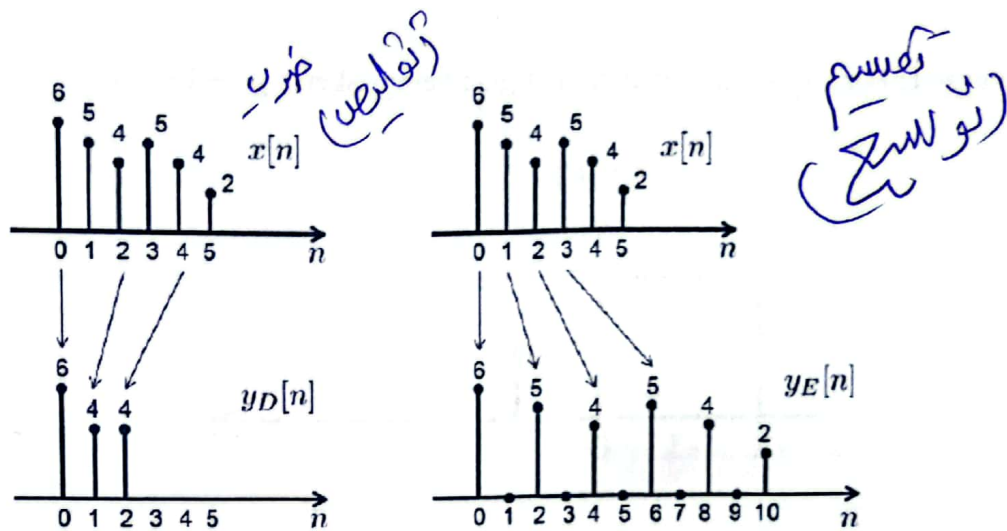
$$y_D[n] = x[Mn], \text{ for some integers } M$$

Where, M is the decimation factor.

Expansion is defined as

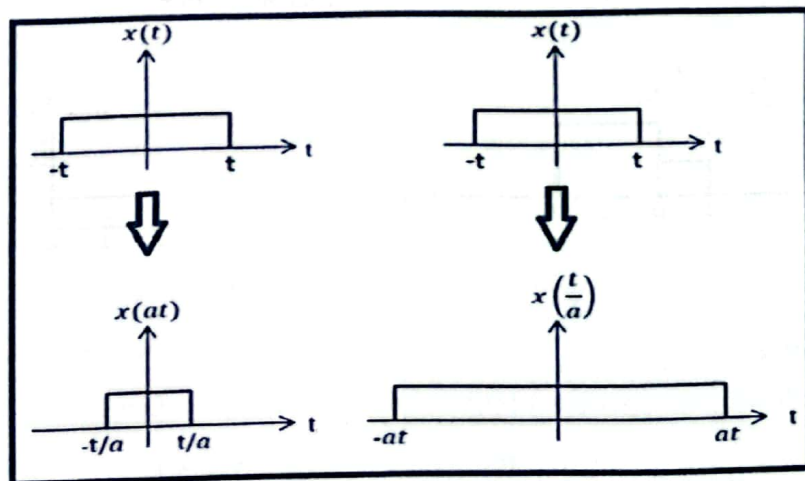
$$y_E[n] = \begin{cases} x\left[\frac{n}{L}\right], & n = \text{integer multiple of } L \\ 0, & \text{otherwise.} \end{cases}$$

Where, L is the expansion factor.



Examples of decimation and expansion for $M = 2$ and $L = 2$.

Example 2

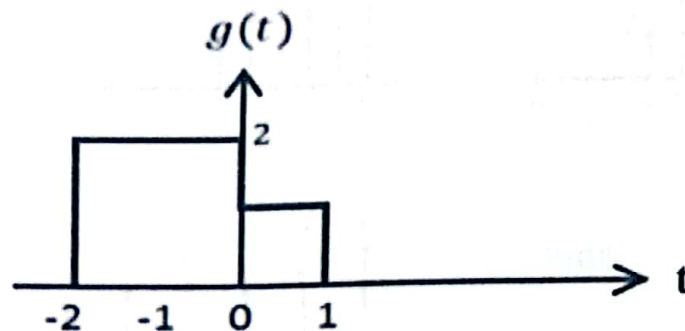


Examples of Time Scaling.

- If $a > 1$, then we have time compression.
- If $a < 1$, then we have time expansion.

Example 1: For the signal shown in the figure below, sketch $g(-1 - \frac{t}{2})$.

$x(t-a)$
 $x(t+a)$



$$\frac{1}{2}((t) - 1)$$

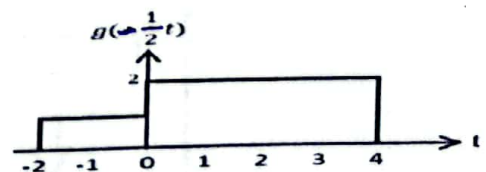
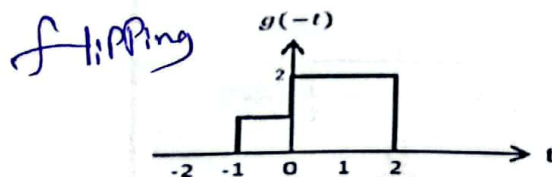
Solution:

$$g\left(-1 - \frac{t}{2}\right) = g\left(-\frac{t}{2} - 1\right) = g\left(-\frac{1}{2}(t+2)\right)$$

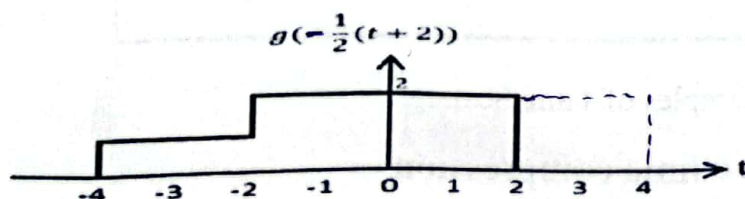
*Handwritten notes: "1 * 2" with an arrow pointing to the 2 in the original expression; "t/2" with an arrow pointing to the t in the original expression; "1/2" with an arrow pointing to the 1/2 in the original expression; "flipping" written near the negative sign in the final expression; "time scaling" written near the 1/2 in the final expression.*

1- Time-inverse

2- Time-scaling by $\frac{1}{2}$



3- shifting to the left by 2



shifting (+2) left

Combination of Operations

Generally, linear operation (in time) on a signal $x(t)$ can be expressed as $y(t) = x(at-b)$. The recommended method is "Shift, then Scale".

Example: The signal $x(t)$ shown in Figure of sketch $x(3t-5)$.

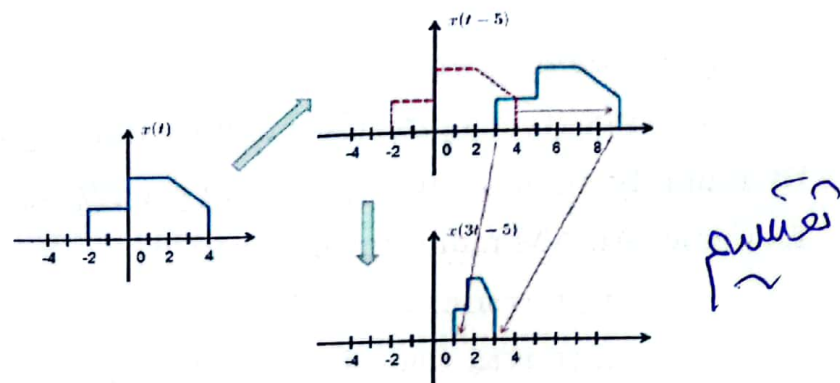


Figure: An example of Shift, then Scale

Operation performed on dependent variable:

Amplitude scaling:

Let $x(t)$ denote a continuous time signal

By amplitude scaling, we get $y(t) = cx(t)$

Where, c is the scaling factor.

Example: An electronic amplifier, a device that performs amplitude scaling.

For discrete time signal $y[n] = cx[n]$

Amplitude addition:

Let $x_1(t)$ and $x_2(t)$ is a pair of continuous time signal

By adding these two signals, we get $y(t) = x_1(t) + x_2(t)$

Example: An audio mixture

For discrete time signal, $y[n] = x_1[n] + x_2[n]$

Amplitude multiplication:

Let $x_1(t)$ and $x_2(t)$ is a pair of continuous time signal

By multiplying these two signals, we get $y(t) = x_1(t) x_2(t)$

Example: An AM radio signal, in which

$x_1(t)$ is an audio signal

$x_2(t)$ is an sinusoidal [carrier wave]

For discrete time signal, $y[n] = x_1[n] x_2[n]$

نقل الإشارة
عن الناقل

AM
FM

Differentiation:

$$y(t) = \frac{d}{dt} x(t)$$

Example: Voltage across an inductor $L \rightarrow v(t) = L \frac{d}{dt} i(t)$

Integration:

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

Example: Voltage across a capacitor $C \rightarrow y(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$

Exponential Signals

A real exponential signal, in its most general form, is written as

$$x(t) = Be^{at},$$

where both B and a are real parameters. The parameter B is the amplitude of the exponential signal measured at time $t = 0$. Depending on whether the other parameter a is positive or negative, we may identify two special cases:

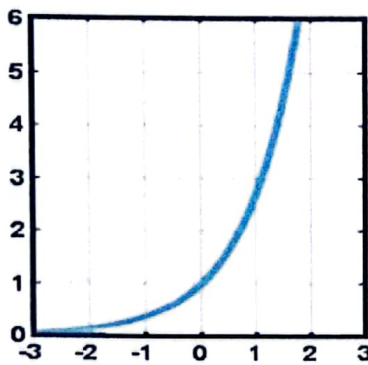
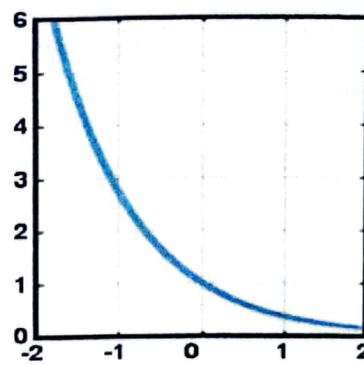


Fig: Growing exponential, for $a > 0$



Decaying exponential, for $a < 0$



In discrete time, it is common practice to write a real exponential signal as $x[n] = Br^n$

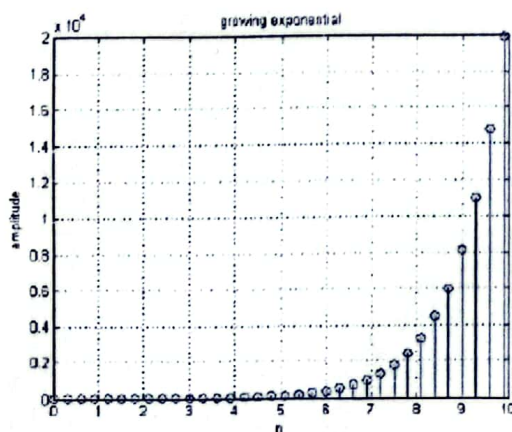
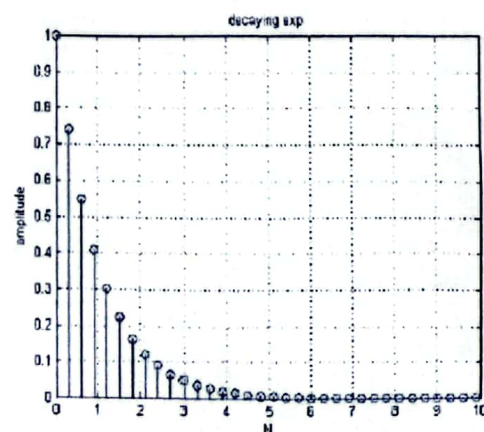


Fig: Growing exponential for $r > 1$

متزايد



Decaying exponential for $0 < r < 1$

متناقص

Assignment

Given signal: $x(t) = \sin(t)$

sketches to show $x(t)$, $x(t-2)$, and $x(2t)$.

describe in your own words how each operation affects the signal.

THANK YOU

2020/04/03